

FAITH, SD, USA

WMO: 726539

Lat: 45.032N

Lon: 102.019W

Elev: 786

StdP: 92.23

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 94056

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.6	-21.2	-27.3	0.3	-23.3	-24.4	0.5	-20.7	15.7	-1.4	14.6	-2.5	5.3	290	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	35.9	19.7	33.4	19.3	31.5	18.9	22.3	29.9	21.3	29.2	20.4	28.3	5.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	16.3	25.4	18.9	15.1	24.5	17.9	14.1	23.7	69.5	30.0	65.5	29.2	62.4	28.1	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	12.7	11.4	-27.3	39.1	2.6	3.1	-29.2	41.3	-30.7	43.2	-32.1	44.9	-34.0	47.2		
(5)			-27.6	24.0	2.5	1.0	-29.4	24.8	-30.9	25.4	-32.3	25.9	-34.1	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-6.2	-6.5	1.0	7.0	12.5	18.6	23.3	21.7	16.7	8.4	1.2	-5.2	(6)	
(7)		DBStd	12.04	7.80	7.86	7.45	5.74	4.90	4.09	3.51	3.76	5.02	5.45	6.50	7.48	(7)	
(8)		HDD10.0	2243	503	461	288	119	30	1	0	0	6	95	268	472	(8)	
(9)		HDD18.3	4221	761	694	537	340	188	45	3	12	88	310	513	730	(9)	
(10)		CDD10.0	1442	0	1	9	29	109	259	414	363	208	46	4	1	(10)	
(11)		CDD18.3	380	0	0	0	1	8	53	159	117	40	2	0	0	(11)	
(12)		CDH23.3	4642	0	0	7	28	135	639	1883	1342	557	53	0	0	(12)	
(13)		CDH26.7	2096	0	0	0	5	33	238	938	636	231	14	0	0	(13)	
(14)	Wind	WSAvg	5.7	6.0	5.6	6.2	6.4	6.0	5.3	5.1	5.1	5.6	5.9	5.7	5.8	(14)	
(15)	Precipitation	PrecAvg	424	7	10	27	48	76	77	58	38	29	26	12	10	(15)	
(16)		PrecMax	704	18	39	112	117	246	153	133	94	91	126	48	24	(16)	
(17)		PrecMin	249	0	0	0	2	23	11	10	8	0	0	0	0	(17)	
(18)		PrecStd	96	5	9	25	33	55	40	32	24	25	28	13	7	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	15.3	24.0	27.3	30.3	35.9	40.2	37.7	36.0	28.6	20.6	13.8	(19)	
(20)			MCWB	5.8	7.7	11.1	13.4	17.4	20.4	20.3	19.4	18.2	14.9	10.4	5.5	(20)	
(21)		2%	DB	8.7	9.6	19.3	22.7	27.0	32.1	36.4	35.1	32.3	23.8	16.2	9.7	(21)	
(22)			MCWB	3.5	4.5	9.3	12.4	16.6	19.5	20.3	19.3	17.5	13.0	8.3	3.6	(22)	
(23)		5%	DB	5.8	6.3	15.7	19.5	24.1	29.1	33.9	32.7	29.2	20.5	13.1	6.9	(23)	
(24)			MCWB	1.9	2.5	7.8	10.6	15.3	18.8	19.9	19.2	16.8	11.3	6.5	2.0	(24)	
(25)		10%	DB	3.7	3.6	11.6	16.7	21.8	27.2	32.1	30.3	26.5	17.3	10.2	4.0	(25)	
(26)			MCWB	0.9	0.8	6.1	9.0	14.0	18.0	19.3	18.6	15.7	10.0	4.8	0.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.6	12.3	14.6	20.1	23.5	24.0	22.9	20.6	15.8	10.7	5.9	(27)	
(28)			MCDB	13.0	14.9	20.4	23.8	26.3	28.8	30.8	31.1	29.2	25.8	20.2	11.8	(28)	
(29)		2%	WB	3.9	4.6	10.0	12.7	18.1	21.5	22.7	21.6	19.2	14.0	8.5	4.0	(29)	
(30)			MCDB	8.3	9.1	18.2	21.7	24.1	28.4	30.3	29.8	27.7	22.0	15.8	9.0	(30)	
(31)		5%	WB	2.2	2.6	8.1	11.1	16.3	20.4	21.7	20.6	18.0	12.0	6.8	2.1	(31)	
(32)			MCDB	5.3	5.9	15.4	18.6	22.5	27.2	29.6	28.6	26.4	18.9	12.7	6.1	(32)	
(33)		10%	WB	1.0	1.1	6.1	9.6	14.8	19.1	20.8	19.7	16.8	10.4	5.2	0.9	(33)	
(34)			MCDB	3.5	3.5	11.5	15.6	20.5	25.5	29.1	27.4	24.7	16.5	9.9	4.2	(34)	
(35)	Mean Daily Temperature Range			MDBR	10.0	10.0	11.4	12.4	12.2	12.8	14.3	14.2	14.0	12.0	11.1	9.3	(35)
(36)		5% DB	MCDBR	12.5	13.0	17.8	18.1	16.3	16.3	17.5	17.8	18.1	17.4	15.2	12.0	(36)	
(37)			MCWBR	8.4	8.4	9.9	9.6	8.1	6.9	5.8	6.4	7.0	8.6	8.6	7.6	(37)	
(38)		5% WB	MCDBR	12.1	12.1	17.2	17.0	14.7	14.5	14.7	15.2	16.0	16.0	14.9	11.8	(38)	
(39)			MCWBR	8.6	8.1	9.8	9.3	8.1	7.2	6.1	6.3	6.9	8.4	8.7	8.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.256	0.261	0.281	0.322	0.343	0.355	0.373	0.367	0.332	0.292	0.263	0.250	(40)	
(41)		taud		2.538	2.535	2.534	2.415	2.426	2.423	2.407	2.401	2.487	2.566	2.550	2.541	(41)	
(42)		Ebn at Noon		888	947	964	941	925	911	890	882	892	894	870	861	(42)	
(43)		Edh at Noon		63	77	89	109	112	113	113	110	93	74	62	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.81	2.83	3.93	5.00	5.94	6.79	6.96	5.88	4.60	2.99	1.99	1.49	(44)	
(45)		RadStd		0.12	0.21	0.31	0.32	0.59	0.48	0.29	0.38	0.30	0.32	0.19	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page